Sanjay Bhandari

University of Utah

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Education

Kalhert School of Computing, University of Utah **August 2025 –**
PhD in Computer Science *Salt Lake City, Utah*

Institue of Engineering, Pulchowk Campus, Tribhuvan University **Oct 2018 – Mar 2023**
Bachelor of Engineering in Computer Engineering *Laltipur, Nepal*

Research Experience

Kalhert School of Computing, University of Utah **Aug 2025 –**
Graduate Research Assistant | Supervisor: Prof. Shireen Elhabian *Salt Lake City, Utah*

- **Statistical Shape Modelling using classical and modern deep learning methods.**
  - Statistical Shape Modelling of Labyrinto and generation of synthetic inner ear anatomies.
  - Spectral embeddings based Variational Autoencoder for synthetic shape generation.

NepAl Applied Mathematics and Informatics Institute **May 2023 – present**
Research Assistant | Supervisor: Dr.Binod Bhattarai *Lalitpur, Nepal*

- **Instance segmentation for Coronary Artery Segmentation and Stenosis Detection**
  - Achieved 2nd position in the ARCADE challenge at MICCAI 2023 despite utilizing robust Convnext-v2 and pseudolabelling to tackle limited data issue.
- **Out-of-Distribution (OOD) Detection in Gastrointestinal Images**
  - Developed a distance-based method for OOD detection in Gastrointestinal setting that utilizes distance information from both nearest and non-nearest clusters, outperforming state-of-the-art methods.
- **Test Time Augmentation for OOD Detection**
  - Created a plug-and-play test time augmentation-based OOD detection method, which significantly improved the performance of all previous OOD detection techniques.
- **Hallucination-Aware Gastrointestinal Multimodal Data Creation and Analysis with Large Vision Language Models**
  - Currently creating a comprehensive multimodal dataset based on the Kvasirv2 dataset to advance VLMs for GI analysis
  - Proposed a metric called GutHal, designed to effectively evaluate whether an LVLM generates hallucinated reports at a fine-grained level and addressing the hallucinations.

NepAl Applied Mathematics and Informatics Institute **Jun 2020 – Apr 2023**
Research Intern | Supervisor: Dr.Binod Bhattarai *Lalitpur, Nepal*

- Minor research activities involving literature review, data labeling, performing experiments, and assisting supervisors in brainstorming research ideas.

Publications

- **Bhandari, S.***, Pokhrel, S.*, Khanal, B.*, … Bhattarai, B., (2025). **Hallucination-Aware Multimodal Benchmark for Gastrointestinal Image Analysis with Large Vision-Language Models**. Medical Image Computing and Computer-Assisted Intervention 2025.[Paper]
- **Bhandari, S.***, Pokhrel, S.*, Ali, S., … Bhattarai, B., (2025). **NCDD: Nearest Centroid Distance Deficit for Out-Of-Distribution Detection in Gastrointestional Vision**. Medical Image Understanding and Analysis 2025.[Paper]
- **Bhandari, S.***, Pokhrel, S.*, Vazquez, E., … Bhattarai, B., (2024). **Cross-Task Data Augmentation by Pseudo-label Generation for Region Based Coronary Artery Instance Segmentation**. Data Engineering in Medical Imaging, MICCAI 2024. [Paper]

- **Bhandari, S.***, Pokhrel, S.*, Vazquez, E., … Bhattarai, B., (2024). **TTA-OOD: Test-time Augmentation for Improving Out-of-Distribution Detection in Gastrointestinal Vision**. Data Engineering in Medical Imaging, MICCAI 2024. **[Paper]**
- **Bhandari, S.***, Pokhrel, S.*, Vazquez, E., … Bhattarai, B., (2023). **Convnextv2 fusion with mask R-CNN for automatic region based coronary artery stenosis detection for disease diagnosis**. arXiv preprint arXiv:2310.04749. **[Paper]**

Professional Experience

Fogsphere (Redev AI Ltd), UK **Jan 2023 – present**
Computer Vision Engineer *Part time, Remote*

- Developed Climbing and Falling detection models for workplace safety through object detection and pose estimation.
- Testing and research optimization for use of ArcFace face recognition model.
- Tracking and Personal Protection Equipment Detection.

Rara Labs, Nepal **Jan 2023 – Mar 2023**
Computer Vision Research Engineer *Remote*

- Worked in anomaly detection in public and private water sources, outlets and reservoirs.

Projects

Laplace Beltrami based Statistical shape modelling of Labyrinto | *Graduate Research Project* **November 2025**

- Statistical Shape Modelling of Labyrinto and generation of synthetic inner ear anatomies
- Used the smallest 'k' eigenmodes of the Laplacian Beltrami operator applied onto labyrinto meshes to model the variation in the inner ear anatomy.

Stenosis Detection and Segmentation| *ARCADE 2023* **[Github]** **November 2023**

- A research project focusing on detection and segmentation of stenosis in coronary arteries.
- Introduced the use of ConvnextV2 as backbone in Mask-RCNN pipeline.

Honors & Awards

- **Named researcher for 35,000 CHF grant on evaluation of Stenosis Detection in Nepal**
- **Runner up of MICCAI 2023 ARCADE challenge among 350 global participants**

Training & Certifications

- Machine Learning and Convolution Neural Networks by Andrew NG (Coursera)
- Nepal Third Winter School in AI (NAAMII)
- Generative Adversarial Networks Specialization(Coursera)
- AWS Academy Graduate - AWS Academy Machine Learning Foundations, Cloud Foundations, Data Analytics

Technical Skills

Languages: Python, C++,Rust
Technologies/Frameworks: Pytorch, Scikit-learn, numpy, mmdetection, , matplotlib, Django
Other tools: Git, LateX, Shell Scripting, Docker, Linux Environment